



Experiments on the water entry of curved wedges: High speed imaging and particle image velocimetry



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ABSTRACT

In this work, we experimentally study the water entry of curved rigid wedges. Experiments are performed on two groups of rigid wedges comprising five specimens each. Each group has a fixed mean deadrise angle (25° and 35°) and varying radius of curvature. Drop tests are conducted in free-fall, and the drop height is parametrically varied to investigate the effect of the entry velocity on the pile-up evolution, the impact dynamics, and the energy transferred to the fluid. Specifically, high speed imaging is utilized to simultaneously measure the penetration depth of the wedge and its wetted surface. Experimental results are used to compute the pile-up coefficient, which is found to be largely independent of both the wedge geometry and the entry velocity, while exhibiting modest variations as a function of the penetration depth. In addition, particle image velocimetry is used to investigate the flow physics generated by the water entry, and especially dissect the energy absorbed by the risen water during impact. Results show that between 60 and 80% of the impact energy is consistently transferred to the risen water, which accounts for the formation of the pile-up and the spray jets.

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1. Introduction

Over the past few decades, considerable research efforts have been devoted to the development of analytical and numerical tools for the prediction of impact dynamics in water entry problems, due to their scientific and technological relevance in naval engineering (Faltinsen, 1990, 2006; Zhao and Faltinsen, 2006; Korobkin, 1996, 2004; Tassin et al., 2010; Kapsenberg, 2011; Abrate, 2013; Hirdaris et al., 2014). While these approaches have been shown to accurately describe hydrodynamic loading in a multitude of experimental studies, their validation is largely limited to specific geometries, whereby the majority of experimental studies have focused on flat wedges (Bisplinghoff and Doherty, 1952; Greenhow, 1987; Engle, 2003; Wu et al., 2004; Carcaterra and Ciappi, 2004; Tveitnes et al., 2008; Lewis et al., 2010; Stenius et al., 2013), cylinders (Greenhow and Yanbao, 1987; Cointe and Armand, 1987; Garrison, 1996; Battistin and Iafrazi, 2003; Sun and Faltinsen, 2006), spheres (De Backer et al., 2009; El Malki Alaoui et al., 2012; Truscott et al., 2012), and cones (De Backer et al., 2009; El Malki Alaoui et al., 2012). More complex geometries, such as paraboloids and pyramids, have only been investigated in theoretical

studies (Scolan and Korobkin, 2001; Korobkin, 2002; Gazzola et al., 2005; Korobkin and Scolan, 2006; Moore et al., 2012), and the effect of geometric curvature on the water entry of rigid bodies is yet to be fully experimentally quantified, see also the review by Abrate (2013).

Modeling the impact dynamics often leverages momentum conservation, whereby the fluid is treated as an added mass whose magnitude increases with time during the water entry. One of the seminal studies in this field is attributed to Von Karman (1929), who formulated an approximate scheme to predict the impact dynamics of a flat wedge. Since then, many authors have extended Von Karman's solution to describe water pile-up in the vicinity of the wedge and its effect on the added mass (Wagner, 1932; Chuang, 1996; Payne, 1995; Wu, 1998; Pesce, 2003; Korotkin, 2009). Central to most of these approaches is the pile-up coefficient, which quantifies the ratio between the horizontal projection of the wetted surface of the wedge and its projected surface with respect to the initial water surface. For a flat wedge, Wagner (1932) predicted that such coefficient equals $\pi/2$, independent of the impact dynamics and deadrise angle. In this context, experimental data by Greenhow (1987) offered evidence that the pile-up coefficient of flat wedges does not vary during impact, even if the wedge decelerates as a result of the fluid–structure interaction. However, Wagner's prediction has been shown to often overestimate the pile-up coefficient of flat wedges (Bisplinghoff and Doherty, 1952) and lose accuracy as the deadrise angle increases (Mei et al., 1999). Notably, the implementation of Wagner condition on

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curved bodies indicates that the pile-up coefficient varies during the impact as a function of both the impact dynamics and the geometry (Faltinsen, 2006). However, a systematic experimental analysis of the pile-up coefficient for curved wedges is currently not available.

As an alternative to momentum conservation, a few modeling efforts rest upon the conservation of energy during impact (Scolan and Korobkin, 2003; Cointe et al., 2004). Importantly, as originally demonstrated by Payne (1993) and recently remarked in a comprehensive review by Abrate (2013), theoretical predictions based on energy conservation may differ from those stemming from momentum conservation, due to the energy imparted to the spray jets. Energy-based formulations typically do not account for the energy imparted to the spray jets, resulting in a discrepancy with respect to schemes that are based on momentum balance. As indicated by Payne (1993), such energy may account for half of the kinetic energy of the wedge, with the remaining half being transformed into the kinetic energy of the added mass. Dissecting the energy of the flow region that is generated as the water is pushed sideways by the falling wedge is critical for accurately implementing energy conservation principles. However, a thorough experimental characterization of the energy transferred to the fluid, along with a detailed experimental analysis of its dependence on the geometry of the wedge, is presently lacking.

In this work, we study the water entry of curved bodies to understand the role of the body curvature on the physics of the impact. Toward this aim, we undertake an extended experimental campaign on wedges with varying radius of curvature entering the water in free-fall from several impact heights, for a total of 40 experimental conditions. We consider both positive and negative curvatures to include convex and concave shapes, spanning the typical spectrum of hull geometries that are considered in the design of marine vessels (Okumoto et al., 2009). High speed imaging is utilized to analyze the impact dynamics of the wedges, in terms of their penetration depth, and the water pile-up phenomenon, quantified through the wetted surface. These experimental results are ultimately used to compute the pile-up coefficient, which is found to be largely independent of the curvature and the penetration depth. We utilize the modified Logvinovich model (Korobkin, 2004) to aid in the interpretation of experimental results and offer further validation for such a well-established approach.

In addition, particle image velocimetry (PIV) (Raffel et al., 2007) is leveraged to investigate the unsteady flow physics of the impact. Specifically, PIV data are used to perform a control volume analysis that is aimed at elucidating the energy transfer from the falling wedge to the fluid. In particular, we seek to isolate the energy absorbed by the fluid bulk from the energy transmitted to the risen water, which is responsible for the pile-up phenomenon and the formation of the spray jets. Results demonstrate that variations of the wedge curvature have negligible influence on the portion of the energy imparted to the risen water. In fact, it is found that the energy imparted to the risen water ranges between 60 and 80% of the total energy transferred to the fluid during the water entry, independent of the curvature and the entry velocity.

The rest of the paper is organized as follows. In Section 2, we introduce the experimental drop-weight apparatus used to perform free-fall impacts, along with the data acquisition system. We also formulate the basic framework of a control volume analysis that is performed to evaluate the energy transferred from the falling body to the risen water during the impact. In Section 3, we report experimental results on the effect of the wedge curvature and the entry velocity on the impact dynamics, on the evolution of the water pile-up, and on the energy transferred to the risen water during the impact. Therein, we also compare experimental results on the pile-up coefficient with theoretical findings obtained by implementing Wagner condition, and we compare analytical predictions based on modified Logvinovich model with the experimental findings. Conclusions and final remarks are summarized in Section 4.

2. Experimental setup and data analysis

In this section, we first present the experimental setup for the drop tests, the data acquisition system, and the design of the specimens. Later, we introduce the details of PIV analyses that are performed to evaluate the fluid kinematics. Finally, we detail the control volume analysis used for the evaluation of the energy transferred to the fluid during the water entry.

2.1. Specimens

In this work, two groups of specimens are studied, see Fig. 1. All the samples have the same square base of width and length $L = 200$ mm. Each group has a fixed mean deadrise angle β_0 of 25° or 35° and comprises five specimens with different radii of curvature R . Such radii of curvature are obtained by varying the deadrise angle at the wedge keel β to attain the values $2\beta_0$, $\frac{3}{2}\beta_0$, β_0 , $\frac{1}{2}\beta_0$, and 0 , according to $R = (L/4)/(\cos\beta_0 \sin(\beta_0 - \beta))$. In the following, we refer to such curvatures with a number from 1 to 5, with 1 identifying the sharpest wedge. Note that curvatures 3 and 5 refer to the special cases of a flat wedge and a cylinder, respectively. Thus, the height is 45 mm and 68 mm for the groups with mean deadrise angle of 25° and 35° , respectively, and the radius of curvature takes the values -131 , -255 , ∞ , 255 , and 131 mm (from curvatures 1 to 5, respectively) for the 25° group, and -106 , -203 , ∞ , 203 , and 106 mm for the 35° group.

Fig. 2 shows the profiles of the ten specimens, where only half of the geometry is shown due to symmetry. Therein, the y -axis is along the specimen height and x is along the width. With respect to this coordinate, the profile of the cross section of the wedge is

$$y = R \left(\cos\beta - \sqrt{1 - \left(\frac{x}{R} + \sin\beta\right)^2} \right) \quad (1)$$

From now onwards, the color coding in Fig. 2 is used to identify the wedges in all the plots; thus, legends in the following graphs are omitted when possible.

All wedges are composed of an internal ABS skeleton printed in a rapid prototyping machine (Stratasys Dimension SST) covered by a 2 mm thick balsa wood layer. Each skeleton has five ribs uniformly placed along its length; the central rib has a thickness of 20 mm and the others of 4 mm. The balsa wood and the ABS skeleton are glued together with an epoxy adhesive with enhanced resistance to water wearing (Loctite E-30CL). Balsa wood is chosen as external layer for its orthotropic properties that ease the process of fully covering the plastic skeleton, while offering significant stiffness to the wedge during impact. A flowable silicone layer is later applied on the outer surface of the wedges to prevent water absorption and the consequent degradation of the specimens. The total mass of each specimen is approximately 0.2 kg.

2.2. Drop weight apparatus and data acquisition

The drop tests are executed on a custom built drop weight machine for the analysis of water entry problems, see also Panciroli and Porfiri (2013, 2014). The apparatus is comprised of a transparent tank 800 mm long, 320 mm large, and 350 mm deep. The tank is

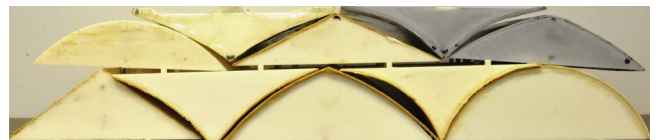


Fig. 1. Curved wedges used in the experimental study; specimens with mean deadrise angle $\beta_0 = 25^\circ$ lie on top of specimens with $\beta_0 = 35^\circ$.

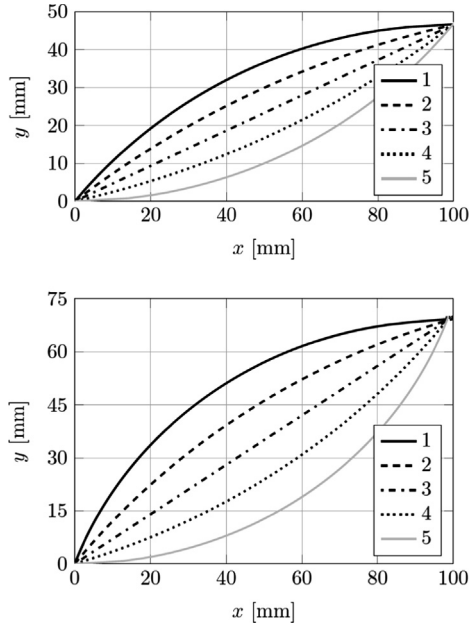


Fig. 2. Cross sections of the specimens used in this study: 25° group (top) and 35° group (bottom). (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

filled up to 300 mm to prevent water overflow after impact. An aluminum frame holds two rails 1.2 m long that are suspended 75 mm above the water surface. A falling sledge runs inside the two rails and its maximum run is 1.1 m. Specimens are rigidly attached to the falling sledge through a metal screw. The total mass of the falling bodies is reported in Table 1. Differences between the masses of the specimens reported in Table 1 are due to variations in the mass of the ABS skeleton, whose ribs maintain the same thickness across geometries. The drop-weight apparatus and the data acquisition system are schematically drawn in Fig. 3.

We consider four impact heights h of 25, 50, 75, and 100 cm for a total of 40 independent conditions. At each impact height and for each specimen, experiments are executed three times, totaling 120 drop tests. Such repetitions are used to ascertain the repeatability of the experiments and highlight possible experimental errors.

A set of markers is drawn on the side of the specimen facing a high speed Dantec Dynamics Phantom SpeedSense 9040 camera. The camera sampling rate is set to 4000 frames per second with a resolution of 640×980 pixels. At time t , the wedge vertical position ξ with respect to the water initial free surface is computed as the average of the vertical displacement of the seven markers. The markers are tracked through the commercial software ProAnalyst (www.xcitex.com). The time t is measured from the beginning of the impact, when the wedge keel touches the water, and we set $\xi(0) = 0$. We filter the entry depth data ξ in time using a moving average over four time instants, corresponding to a window of 1 ms. The entry velocity $\dot{\xi}$ is evaluated through differentiation of the position using a central difference approximation.

A second identical high speed camera is positioned alongside of the first camera, but centered to acquire only one-half of the body and observe the evolution of the water pile-up. Note that the spatial resolution is inverted with respect to the first camera, which is used to track the vertical displacement of the wedge, rather than the pile-up evolution. The two cameras are synchronized to acquire the impact simultaneously. A custom-made code in Matlab (www.mathworks.com) is utilized to manually locate the last wetted point of the wedge at each time step, corresponding to the root of the water jet that is formed along the impacting wedge and ends up as a spray (Faltinsen, 2006). The accuracy of such estimation is within 0.5 mm. The analysis

Table 1

Total mass of the ten falling bodies used in the experiments, including the specimens and the sledge.

Curvature	$\beta_0 = 25^\circ$ (kg)	$\beta_0 = 35^\circ$ (kg)
1	0.361	0.366
2	0.381	0.395
3	0.391	0.412
4	0.386	0.433
5	0.426	0.445

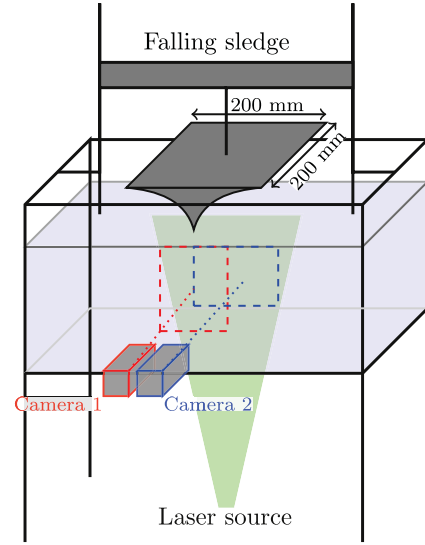


Fig. 3. Sketch of the experimental setup with overlaid specimens characteristic dimensions, laser sheet location (used for PIV analysis), and position and relative field of view of the two cameras.

is terminated when the wedge is fully submerged or the location of the root of the water jet is not detectable from the images, due to the wedge motion outside the field of view or water splash. To study the pile-up phenomenon, both the vertical and horizontal displacement of the root of the water jet point are averaged across the three tests that are performed for each impact condition. Data are also smoothed in time using a moving average over four points.

Fig. 3 also shows a laser source located below the water tank and pointing upward to illuminate the wedge at its mid-span during the PIV analysis, whose details are presented in Section 2.3. PIV experiments are executed independent of the 120 tests mentioned earlier, due to the different illumination conditions required in the analysis. Specifically, PIV experiments are conducted in the dark, while the impact dynamics is studied in laboratory illumination. For each condition, PIV experiments are executed once for a total of 40 PIV tests.

The friction in the rails and the air resistance are responsible for a slight reduction of the entry velocity with respect to the nominal value $\sqrt{2gh}$, where g is the gravity acceleration. Such reduction is consistently on the order of 10%. Table 2 shows the mean entry velocity $\bar{v}$ and the standard deviation σ_v that are defined by

$$\bar{v} = \frac{1}{n} \sum_{i=1}^n v_i \quad (2)$$

and

$$\sigma_v = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (v_i - \bar{v})^2} \quad (3)$$

where $n=30$ is the number of trials for each impact height.

2.3. PIV measurements

PIV is utilized to study the fluid flow during the impact. Images are captured by the second high speed camera (Camera 2 in Fig. 3) at 4000 frames per second with a resolution of 960×480 pixels. Hollow glass spheres of $44 \mu\text{m}$ mean diameter are used as seeding. The camera is located 60 cm away from the laser sheet and is centered to capture only half of the wedge, due to the expected symmetry with respect to the vertical axis. A 5W Nd:YAG Ray-power continuous laser with thickness of approximately 2 mm is located at the bottom of the water tank pointing upwards to illuminate the water at the mid-span of the wedge, where three-dimensional effects are expected to be minimized.

Fig. 4 displays an example of a high speed image with overlaid velocity vectors and contour lines computed through PIV analysis. Note that PIV analysis is restricted to the fluid bulk region $\mathcal{R}_1$, due to practical challenges in resolving the flow physics in the smaller domain of risen water $\mathcal{R}_2$ adjacent to the water surface, where light reflections are stronger. Specifically, $\mathcal{R}_1$ identifies the fluid domain in the camera field of view which is below the pristine water surface, while $\mathcal{R}_2$ includes the pile-up region and the spray jet, generated as the water is pushed sideways by the falling body, see also the description by Howison et al. (1991). As detailed in what follows, energy conservation is utilized to estimate the energy transferred to $\mathcal{R}_2$, whose analysis from PIV data would otherwise be unfeasible.

In the PIV analysis, a moving mask is used to exclude the portion of the image covered by the wedge that progressively enters the water. The masking process is automated through a Matlab script that takes as input the experimental data of the impact dynamics and the wedge geometry. Specifically, the impact dynamics of the least noisy of the corresponding three repetitions presented in Section 2.2 is considered in this post-processing phase. PIV analysis is performed using the open-source Matlab code PIVlab (Thielicke and Stamhuis, 2010). This analysis computes the instantaneous velocity field at the mesh points in a regular grid pattern. A multigrid approach (Scarano and Rie-

thmuller, 1999, 2000) with decreasing interrogation area size of 64×64 , 32×32 , and 16×16 pixels and 50% window overlap is utilized, and the velocity field is resolved through a FFT-based cross-correlation technique. A 2×3 Gaussian subpixel interpolation scheme is employed for subpixel resolution.

2.4. Computation of the energy transferred to the water pile-up during impact

Assuming that viscosity and friction are negligible and that the impact is adiabatic, we perform a control volume analysis in a fixed domain enclosing the whole falling wedge. By neglecting three-dimensional phenomena, energy conservation in the control surface $\mathcal{R}^*$ yields (Welty et al., 2007)

$$\dot{\mathcal{E}} + \frac{d}{dt} \int_{\mathcal{R}} \rho e d\mathcal{R} = - \int_{\Gamma} \rho e (\mathbf{v} \cdot \mathbf{n}) d\Gamma - \dot{W} \quad (4)$$

where $\mathcal{E}$ is the total energy of the falling body; e is the specific energy of the fluid; ρ is the constant fluid density equal to $\rho = 1000 \text{ kg/m}^3$; $\mathcal{R}$ is the fluid domain enclosed by $\mathcal{R}^*$; Γ is the portion of the perimeter of $\mathcal{R}^*$ in the fluid domain; $\mathbf{n}$ is the unit outward normal to Γ ; $\mathbf{v}$ is the water velocity; and $\dot{W}$ is the rate of work per unit wedge length (along the wedge axis that is orthogonal to the plane in Fig. 4) done by the fluid in $\mathcal{R}$ on the rest of the fluid.

Each of the terms in Eq. (4) can be expressed in terms of the wedge motion and the fluid pressure and velocity. Specifically, the energy per unit length of the falling wedge is

$$\mathcal{E} = \frac{1}{2} M \dot{\xi}^2 - Mg\xi \quad (5)$$

where M is the dry mass per unit length. Moreover, the specific energy of the fluid is

$$e = \frac{1}{2} \|\mathbf{v}\|^2 + gy \quad (6)$$

where y is aligned with the vertical axis. Finally, the power per unit length spent by the fluid in $\mathcal{R}$ on the surrounding fluid is

$$\dot{W} = \int_{\Gamma} p (\mathbf{v} \cdot \mathbf{n}) d\Gamma \quad (7)$$

where p is the gauge pressure.

Due to symmetry, we consider only half of the fluid domain. In addition, we partition the halved fluid domain into $\mathcal{R}_1$ and $\mathcal{R}_2$, see Fig. 5. Thus, upon substituting Eqs. (5)–(7) in Eq. (4), we express the rate of change of the energy in $\mathcal{R}_2$ as

$$\begin{aligned} \frac{d}{dt} \int_{\mathcal{R}_2} \left(\frac{1}{2} \rho \|\mathbf{v}\|^2 + \rho gy \right) d\mathcal{R} = & - \frac{d}{dt} \int_{\mathcal{R}_1} \left(\frac{1}{2} \rho \|\mathbf{v}\|^2 + \rho gy \right) d\mathcal{R} \\ & - \int_{\Gamma_1 \cup \Gamma_2} \left(\frac{1}{2} \|\mathbf{v}\|^2 + gy + \frac{p}{\rho} \right) \rho (\mathbf{v} \cdot \mathbf{n}) d\Gamma - \frac{1}{2} \frac{d}{dt} \left(\frac{1}{2} M \dot{\xi}^2 - Mg\xi \right) \end{aligned} \quad (8)$$

where Γ_1 and Γ_2 are the portions of Γ indicated in Fig. 5. Note that the control surface is considered to be sufficiently large to contain the spray sheets for the whole duration of the PIV acquisition, so that $\dot{W}$ is computed only on the basis of contributions in the fluid bulk.

The velocity field in the first two terms on the right-hand side of Eq. (8) is computed from PIV data. The estimation of the pressure field in the second term in the right-hand side of Eq. (8) is performed following the methodology proposed in Panciroli and Porfiri (2013), which is based on the integration of Navier–Stokes equations from PIV data. However, different from Panciroli and Porfiri (2013), we account for the presence of body forces (gravity) in the computation of the pressure at the boundaries of the control domain. All the spatial integrations are performed using the space-marching algorithm proposed in Baur and

Table 2
Mean entry velocity $\bar{v}$ and standard deviation σ_v for all the impact tests at the impact height h .

h (cm)	$\bar{v}$ (m/s)	σ_v (m/s)
25	1.87	0.11
50	2.77	0.14
75	3.38	0.12
100	3.82	0.22

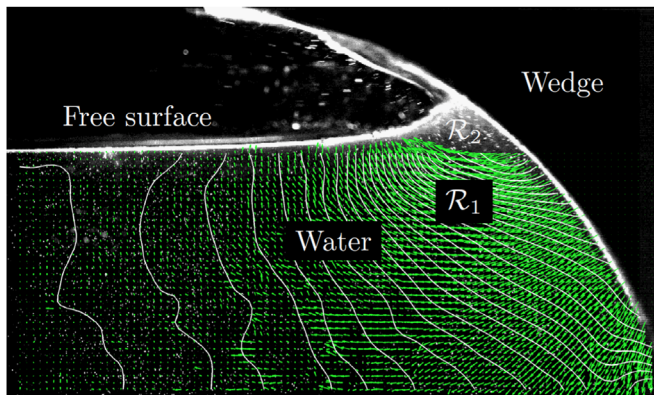


Fig. 4. Example of a high speed image with overlaid PIV measurement and relative contour lines. The wedge considered here has an average deadrise angle $\beta_0 = 35^\circ$ and deadrise angle at the keel of $2\beta_0$.

Table 4

Maximum percent error between the three trials of the tests for wedges with average deadrise angle of 35° .

Curvature	h			
	25 cm	50 cm	75 cm	100 cm
1	3.40	4.28	2.33	0.91
2	1.96	1.55	3.44	2.24
3	1.82	1.46	1.14	1.30
4	0.90	0.56	0.58	2.79
5	0.70	2.31	0.26	1.43

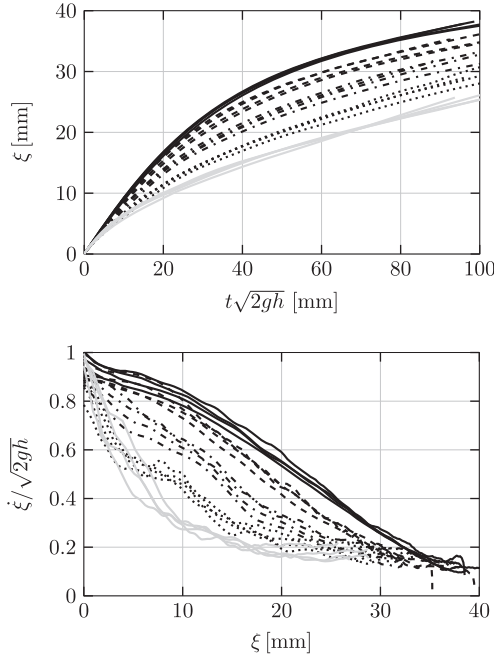


Fig. 7. Collection of experimental results for wedges with 25° average deadrise angle. Entry depth versus impact time multiplied by the nominal entry velocity $\sqrt{2gh}$ (top) and entry velocity scaled by the nominal entry velocity $\sqrt{2gh}$ versus entry depth (bottom). A single test is presented per condition.

the entry depth are correlated to the entry velocity, whereby experiments on the same sample for different heights coalesce on the same curve after scaling, as shown in the top panels of Figs. 7 and 8. Similarly, as displayed in the bottom panels of Figs. 7 and 8, the plots of the normalized entry velocity $\xi/\sqrt{2gh}$ versus the entry depth are also found to collapse on the same curve for different drop heights.

These findings indicate that the impact of the cylindrical wedge is characterized by a strong reduction of the velocity in the early stage of the impact. On the other hand, sharper wedges display a velocity reduction that is milder at the beginning of the impact, while increasing with the entry depth. Notably, the wedges with initial deadrise angle equal to $\frac{3}{2}\beta_0$ (curvature 2) produce a velocity that decreases almost linearly with the entry depth.

3.3. Analysis of the pile-up during water entry

Here, we evaluate the influence of the curvature and the entry velocity on the evolution of the pile-up. Fig. 9 shows a representative high speed image used for the analysis with overlaid notations. In addition to the entry depth ξ , Fig. 9 illustrates the wetted length r^* and the reference wetted length r . The latter quantity can be written in terms of the entry depth ξ and

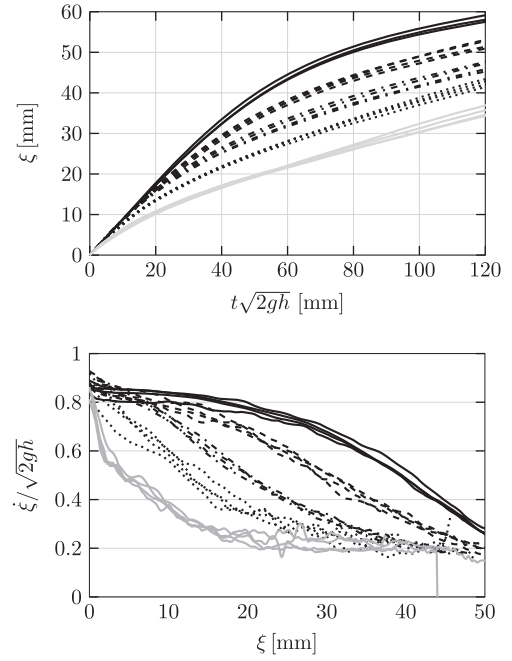


Fig. 8. Collection of experimental results for wedges with 35° average deadrise angle. Entry depth versus impact time multiplied by the nominal entry velocity $\sqrt{2gh}$ (top) and entry velocity scaled by the nominal entry velocity $\sqrt{2gh}$ versus entry depth (bottom). A single test is presented per condition.

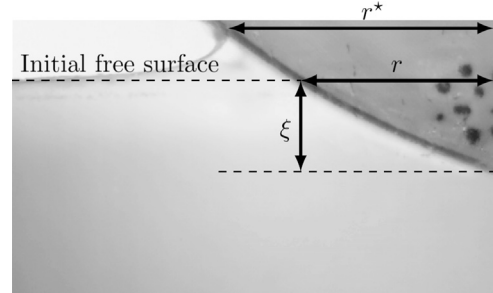


Fig. 9. Example of a high speed image captured by Camera 2 during the water entry of a wedge with average deadrise angle $\beta_0 = 35^\circ$ and deadrise angle at the keel of $\frac{1}{2}\beta_0$ (curvature 3) with overlaid notations.

geometry of the wedge by using Eq. (1), namely

$$r = R \left(-\sin \beta + \sqrt{1 - \frac{(R \cos \beta - \xi)^2}{R^2}} \right) \quad (10)$$

Notably, for the limit cases of a flat wedge and a cylinder, $r = \xi / \tan \beta$ and $r = R \sqrt{1 - (R - \xi)^2 / R^2}$, respectively.

Fig. 10 shows the pile-up coefficient $\phi_r = r^*/r$ quantifying the water pile-up phenomenon as a function of the entry depth ξ . The number of frames N^* considered in these plots varies with the condition, as the analysis of the water pile-up is terminated at different times depending on the experiment. Importantly, the family of plots nearly coalesces on the same curve with respect to the entry depth. In agreement with our expectations based on the literature (Dobrovolskaya, 1969; Greenhow, 1987; Mei et al., 1999), experimental results suggest that the pile-up coefficient ϕ_r is lower than the value of $\pi/2$ predicted by Wagner (1932) for flat wedges, independent of the mean and the local deadrise angle. Such coefficient varies between 1.1 and 1.4, and slightly increases with the entry depth. Note that the plots show a smaller range of

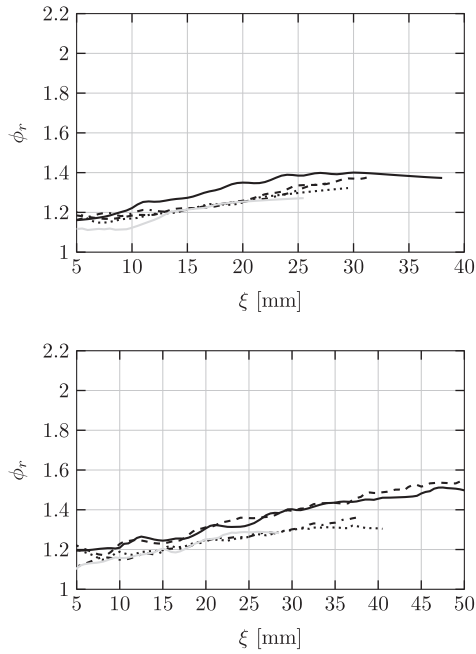


Fig. 10. Normalized horizontal projection of the wet length versus the entry depth for wedges with mean deadrise angle $\beta_0 = 25^\circ$ (top) and $\beta_0 = 35^\circ$ (bottom), and the largest drop height of 100 cm. The computation of the pile-up coefficient is based on the average of the three repetitions executed for each condition.

Table 5

Mean coefficients $\bar{\phi}_r$ and maximum values of σ_r for all the wedge shapes.

Wedge mean deadrise angle		Curvature				
		1	2	3	4	5
25°	$\bar{\phi}_r$	1.33	1.25	1.23	1.23	1.21
	$\max \sigma_r$	0.06	0.09	0.08	0.07	0.04
35°	$\bar{\phi}_r$	1.34	1.34	1.23	1.25	1.23
	$\max \sigma_r$	0.10	0.12	0.09	0.05	0.04

entry depth than Figs. 7 and 8, which also include time frames where the wedge is entirely submerged.

To quantify variations of the pile-up coefficient with respect to the experimental condition, we compute

$$\sigma_r = \frac{\phi_r}{\bar{\phi}_r} - 1 \quad (11)$$

where $\bar{\phi}_r$ is the mean value of the pile-up coefficient computed by averaging the time evolution of such coefficient over the four drop heights, for each curvature and mean deadrise angle. Table 5 reports the values $\bar{\phi}_r$ and the maximum values of the coefficient σ_r for all the specimens. Results show that the maximum variation between the pile-up coefficient and its mean value never exceeds 12% for all the curvatures, with values as low as 4%.

These findings may support the possibility of relating the horizontal projection of the wetted length r^* of the falling wedge to its reference value r through a universal scaling coefficient that is approximately equal to 1.27 (computed as the mean value of $\bar{\phi}_r$ across all experimental conditions). Such coefficient is thus independent of the local deadrise angle, the curvature of the wedge, and the deceleration during the impact.

For the sake of clarity, results in Fig. 10 are drawn for the higher impact heights only, identifying the most severe impact cases.

However, no effect of the entry velocity is found, and curves obtained for the other entry velocities follow similar paths.

3.4. Analysis of the impact dynamics using the modified Logvinovich model

To further elucidate the effect of the curvature and the entry velocity on the evolution of the pile-up and impact dynamics, we utilize the modified Logvinovich model (Korobkin, 2004). Within this framework, neglecting surface tension, gravity, viscosity, and compressibility effects, the pressure distribution on the wetted length is expressed as (Korobkin, 2004)

$$p_a(x, t) = \rho \xi_a^2(t) \left[\frac{\dot{r}_a^*(t)}{\xi_a^*(t)} \frac{r_a^*(t)}{\sqrt{r_a^{*2}(t) - x^2}} - \frac{1}{2} \frac{r_a^{*2}(t)}{r_a^{*2}(t) - x^2} \frac{1}{1 + (y'(x))^2} - \frac{1}{2} \frac{(y'(x))^2}{1 + (y'(x))^2} + \frac{\ddot{\xi}_a(t)}{\xi_a^2(t)} \left(\sqrt{r_a^{*2}(t) - x^2} + y(x) - \xi_a(t) \right) \right] \quad (12)$$

where the subscript “a” is used to identify analytical results and a prime indicates differentiation with respect to x . In Equation (12), r_a^* is calculated from Wagner condition, which is enforced through (Korobkin, 1996, 2004)

$$\int_0^{\frac{\pi}{2}} y(r_a^*(t) \sin \theta) d\theta - \frac{\pi}{2} \xi_a(t) = 0 \quad (13)$$

The hydrodynamic force exerted on the wedge is obtained by integrating the pressure along the wetted length as (Korobkin, 2004; Korobkin and Malenica, 2005)

$$F_a(t) = 2 \int_0^{r_a^*(t)} \max\{0, p_a(x, t)\} dx \quad (14)$$

Finally, by neglecting gravity, the impact dynamics can be predicted from Newton's second law through

$$m \ddot{\xi}_a(t) = -F_a(t) \quad (15)$$

where m is the total dry mass of the falling wedge.

We integrate the system using the fourth order Runge–Kutta method (Press et al., 1993) with initial conditions $\xi_a(0) = 0$ and $\dot{\xi}_a(0) = \bar{v}$ (with reference to Table 2). To march from time t_0 to time $t_0 + \Delta t$, where Δt is the time increment, we follow this series of algorithmic steps: (i) we numerically integrate Eq. (13) and its time derivative using Gauss quadrature points (Press et al., 1993) and solve the resulting nonlinear algebraic equations to compute $r_a^*(t_0)$ and $\dot{r}_a^*(t_0)$; (ii) we estimate the pressure field from Eq. (12) using $r_a^*(t_0)$, $\dot{r}_a^*(t_0)$, $\xi_a(t_0)$, $\dot{\xi}_a(t_0)$, and $\ddot{\xi}_a(t_0)$ and from there we identify the range $I(t_0) \subset [0, r_a^*(t_0)]$ where such pressure is positive; (iii) we use Gaussian quadrature in $I(t_0)$ to write the right-hand side of Eq. (15) as a function $\xi_a(t)$, $\dot{\xi}_a(t)$, and $\ddot{\xi}_a(t)$ for all t between t_0 and $t_0 + \Delta t$; and (iv) we express such second order equation as a system of two coupled first order equation and we integrate using Runge–Kutta from t_0 and $t_0 + \Delta t$. The time step in the integration scheme is set to $\Delta t = 0.05$ ms and the number of quadrature points used in all the numerical integrations is 40.

The pile-up coefficient is computed consistently with the experimental analysis as r_a^*/r , for $r_a^* < L/2$. Fig. 11 displays analytical results for the pile-up coefficient for all the experimental conditions addressed in this study. At the onset of the impact, the pile-up coefficient tends to $\pi/2$ for all the geometries except of the cylinder (gray lines), in which case the initial value is $\sqrt{2}$, as one can compute by taking a series expansion of Wagner condition for small penetration depths following Faltinsen (2006). As expected from observations in Bisplinghoff and Doherty (1952) and Mei et al. (1999), the pile-up coefficient is generally overestimated with respect to experimental results in Fig. 10. In agreement with experimental results, theoretical predictions suggest that the pile-up coefficient is

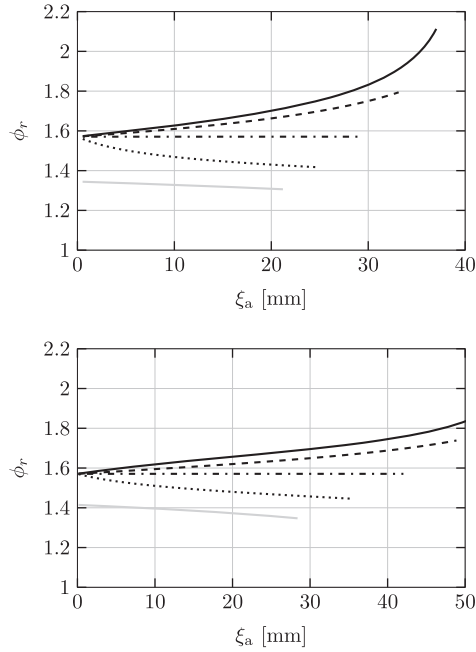


Fig. 11. Pile-up coefficient computed from Wagner condition for wedges with mean deadrise angle $\beta_0 = 25^\circ$ (top) and $\beta_0 = 35^\circ$ (bottom).

smaller for cylindrical geometries, even if theoretical results suggest a more marked reduction than experimental data. Wagner condition is also successful in anticipating the moderate increase in the pile-up coefficient as a function of the entry depth observed in experimental data.

Fig. 12 shows analytical predictions for the normalized entry velocity versus the entry depth until $r_a^* < L/2$. The solution is in very good agreement with experimental results displayed in Figs. 7 and 8, whereby the error between experimental findings ξ and analytical predictions ξ_a , defined as

$$\sqrt{\frac{1}{\min\{N^*, N_a^*\}} \sum_{i=1}^{\min\{N^*, N_a^*\}} \frac{(\xi(t_i) - \xi_a(t_i))^2}{\xi_a(t_i)^2}} \quad (16)$$

is lower than 6% for almost all the wedges and entry velocities, except for the cylinder of the 25° family, for which the error reaches 22.22% for the impact height of 50 cm. Here N_a^* corresponds to the frame for which r_a^* exceeds $L/2$. For the drop height of 25 cm, the maximum error across all geometries is only 4.77%, and for the drop height of 100 cm it is less than 12%. We should also acknowledge that the cylinder falling from 50 cm is the condition for which we noted the highest variation between the test repetitions, as shown in Table 3.

For illustration purposes, in Fig. 13, we report the acceleration as a function of time for the representative case of an impact height of 50 cm. Results displayed therein demonstrate that the largest acceleration is attained by cylinders, whereas sharper wedges exhibit lower accelerations and peak value is attained later in the impact. As commented in Zhao et al. (1996), three-dimensional effects at the edges of the body might influence the total hydrodynamic force as the entry depth increases. For the unitary width-to-length ratio of the bodies used here, three-dimensional effects are expected to reduce the total hydrodynamic loading by approximately 20% when the body is completely wetted (Zhao et al., 1996). In the early phase of the impact, until the entry depth reaches 15 mm, such reduction should only amount to 5% (Zhao et al., 1996).

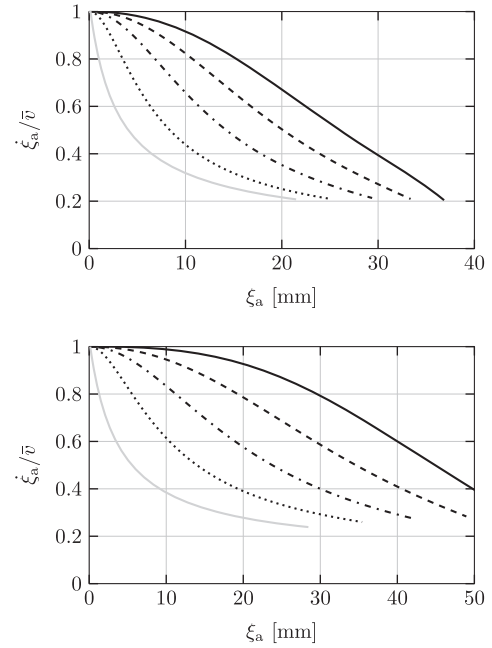


Fig. 12. Analytical predictions for the normalized entry speed versus the entry depth. Curved wedges with average deadrise angle 25° (top) and 35° (bottom). Normalized predictions are identical for all impact heights.

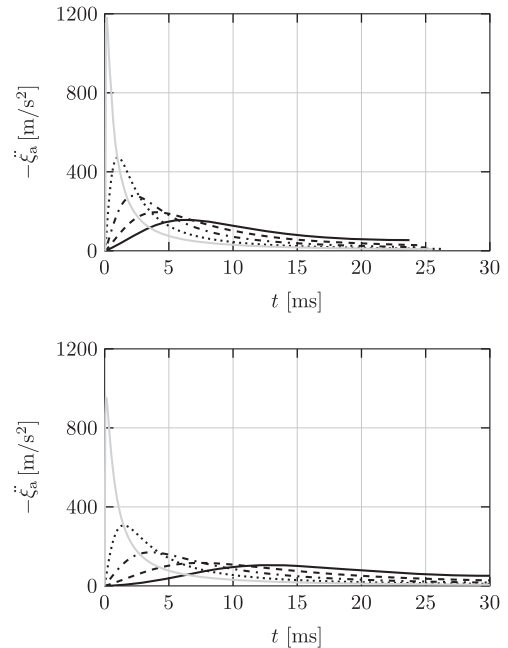


Fig. 13. Analytical solution for the time evolution of the acceleration of wedges with average deadrise angle of 25° (top) and 35° (bottom) and different curvatures. Results are for an entry velocity of 2.77 m/s, corresponding to an impact height of 50 cm, see Table 2.

3.5. Analysis of the energy transferred to the fluid

Here, we analyze the energy transferred to the risen water region $\mathcal{R}_2$, comprising the water pile-up and the jet spray, computed through the control volume analysis in Section 2.4. Specifically, in Fig. 14, we report the energy transferred to the fluid during the water entry as a fraction of the initial energy of the wedge $(\mathcal{E}(0) - \mathcal{E}(t))/\mathcal{E}(0)$ and, in Fig. 15, we dissect the energy contribution in the inner flow region as a function of the entry depth. We note that $\mathcal{E}(0)$ and $\mathcal{E}(t)$ in Figs. 14 and 15 are calculated

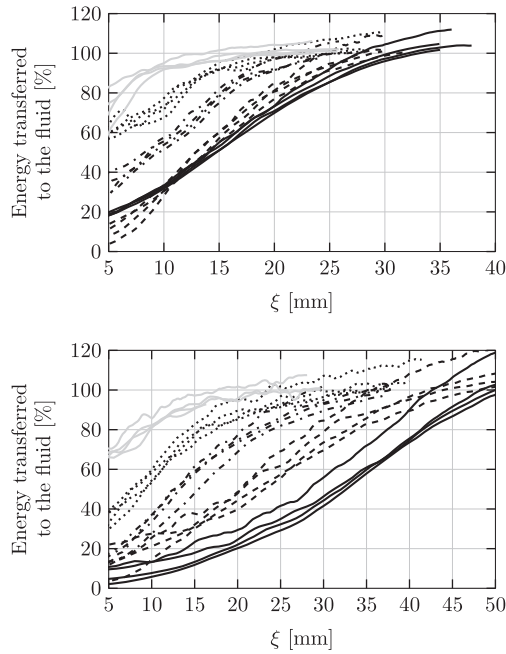


Fig. 14. Energy transferred to the fluid $(\mathcal{E}(0) - \mathcal{E}(t))/\mathcal{E}(0)$ as a function of the entry depth. Results are for wedges with average deadrise angle of 25° (top) and 35° (bottom) until the wedge is fully submerged.

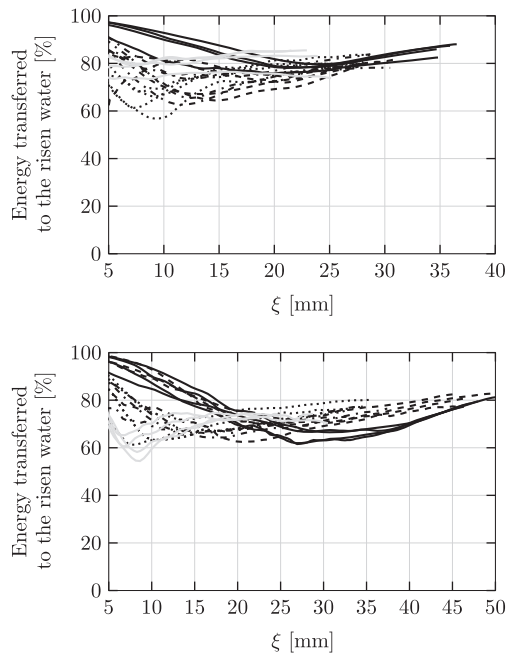


Fig. 15. Energy imparted to $\mathcal{R}_2$ as a fraction of the total energy transferred to the fluid as a function of the entry depth. Results are for the wedges with average deadrise angle of 25° (top) and 35° (bottom). Percent values are computed by taking the ratio between the energy in $\mathcal{R}_2$ and half of the energy transferred to the fluid, which is depicted in a normalized form in Fig. 14.

for each condition (average of three repetitions), not from the aggregated data in Table 2. Values above 100% in Fig. 14 should be related to uncertainties in the estimation of the wedge motion, entry depth and velocity, from images.

Results in Fig. 15 show that the fraction of the energy transferred to the risen water is almost constant over the whole duration of the impact for cylinders (gray line). Such energy approximately settles between 60 and 80% of the total energy imparted to the fluid after a penetration less than 10 mm. Sharper

wedges display a smoother dependence of the fraction of energy imparted to $\mathcal{R}_2$, while the terminal value is highly comparable to results for cylinders. In addition, our results support the hypothesis that the energy balance between $\mathcal{R}_1$ and $\mathcal{R}_2$ is independent of the entry velocity. These findings are in good agreement with the prediction of Scolan and Korobkin (2003), suggesting that a significant portion of the energy is imparted to $\mathcal{R}_2$.

Three-dimensional effects are expected to influence the PIV analysis underpinning the estimation of the energy transferred to $\mathcal{R}_2$ region only in the second half of the impact, when the entry depth reaches 15 mm for the 25° group and 20 mm for the 35° group. However, such effects are not anticipated to significantly influence the overall energy balance in the fluid for most of the conditions, whereby we notice that for all the wedges with curvatures 3, 4, and 5, more than 70% of the total energy transferred to the fluid is imparted during the first half of the impact, when the flow can be accurately described as two-dimensional. For the remaining wedges (with curvatures 1 and 2), the energy transferred to the fluid during the first half of the impact is lower than 60% of the total energy transferred to the fluid, indicating that three-dimensional phenomena may be relevant. Future studies will focus on the use of stereoscopic PIV (Prasad, 2000) to systematically dissect the effect of three-dimensional phenomena in water entry problems, including also bodies with double curvatures.

Finally, we comment that PIV analysis is likely to underestimate the velocity field in the fluid closer to the wetted surface of the wedge, especially in the early stage of the impact. In fact, this region is characterized by a high velocity gradient, which may be underestimated by PIV. Such effects can be partially alleviated through the selection of the control area as discussed in Panciroli and Porfiri (2013). For illustration, we consider as reference case the water entry of the cylinder with mean deadrise angle of 35° and largest drop height of 100 cm, which is characterized by strong acceleration at the onset of the impact, see Fig. 13. By shrinking $\mathcal{R}_1$ along its diagonal until $\mathcal{I}_2$ touches the wedge, we find that variations in the energy contributions are less than 4% for the whole duration of the impact.

4. Conclusions

This work presents the results of an experimental study on the water entry of curved bodies, which was designed to investigate the role of the curvature on the impact dynamics, the water pile-up evolution, and the energy imparted to the fluid during water entry. Experiments were performed on two families of curved wedges with fixed mean deadrise angle (25° and 35°) and five different curvatures, including the limit cases of flat wedges and cylinders. Experiments were conducted in free-fall and a total of 40 experimental conditions were studied. High speed imaging and PIV were integrated to establish a thorough understanding of both the kinematics of the fluid flow and its kinetics during the water entry.

Our results indicate that, during the impact, the pile-up coefficient associated with the horizontal projection of the wetted surface is approximately independent of the entry speed, the mean curvature, and the curvature at the keel. This pile-up coefficient is approximately 1.27. Such experimental findings are lower than the predictions of the pile-up coefficient calculated from Wagner condition. We have utilized the modified Logvinovich model to predict the impact dynamics of the curved wedges and found very good agreement between experimental results and theoretical predictions. Our findings on the flow kinetics demonstrate that between 60 and 80% of the energy transferred to the fluid during the impact is imparted to the risen water, which accounts for the pile-up and the spray jets, independent of the mean curvature, the curvature at the keel, and the entry velocity.

Findings of this study can be used for the validation of modeling schemes based on both momentum and energy conservation, by offering a direct characterization of the pile-up phenomenon as a function of the impact dynamics, along with a detailed analysis of the energy transfer to the risen water. Such modeling schemes are critical for the structural design of ship hulls, whose geometric curvature influences the stability and resilience of the marine vessel (Okumoto et al., 2009). In addition, our results are expected to aid in the validation of theoretical and numerical schemes that seek to describe the complex flow physics generated during water entry, by offering, for the first time, a thorough characterization of the flow kinematics and kinetics for a wide set of wedge geometries.

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